# Roberto Gallego Barbarán

Junior Software Engineer | Applied AI & Backend
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## PROFESSIONAL SUMMARY

Junior software engineer focused on applied AI and backend, currently an AI & Automation intern at DEKRA and finishing a BSc in Software Engineering (University of Málaga, 2027). Build LLM agents in Python (LangChain, LangGraph, MCP) and backend services in Go and Java. 4th place in the Betsson challenge at Hackathon UMA 2026 (300+ students).

## TECHNICAL SKILLS

**Languages:** Python, Go, Java 17, TypeScript, JavaScript, SQL.
**Applied AI:** LangChain, LangGraph, MCP, smolagents, computer vision/OCR.
**Backend & APIs:** Spring Boot, Spring MVC, Spring Security, JPA, REST, gRPC.
**Distributed ledgers:** Canton (Ledger API v2), Daml.
**Databases:** MariaDB/MySQL.
**Cloud & infrastructure:** Docker, Docker Compose, Nginx, Linux, Hugging Face Spaces, OpenTelemetry.
**Tools & practices:** Git, GitHub Actions (CI/CD), Maven, Playwright, pytest, ruff, JUnit, Mockito.

## PROFESSIONAL EXPERIENCE

**DEKRA** Sept. 2026 – Mar. 2027
*AI & Automation Engineer Intern* Málaga, Spain

- Build AI agents in Python with LangChain and LangGraph to automate the analysis of laboratory results.
- Connect LLMs to external tools through MCP and orchestrate decision flows with LangGraph state graphs.

**IOBuilders (University-Industry Collaboration)** May 2026 – Jul. 2026
*Backend Developer (Canton/DLT)* Málaga, Spain

- Built the Go integration layer between the REST backend and the Canton Ledger API v2 (team of three): a shared client package reused by the API and the event listener, using gRPC with an HTTP JSON API fallback.
- Delivered Delivery-versus-Payment (DvP) settlement end to end (create, execute, cancel and withdraw endpoints, UI and API docs) on a three-participant Dockerized Canton network.
- Added real-time ledger monitoring with a Go listener and OpenTelemetry observability (OTLP collector, .NET Aspire dashboard), and fixed a race condition in contract-factory initialization.

## PROJECTS

**Cava Agent** | *Python, smolagents, Gradio, pytest, GitHub Actions* | GitHub 2026

- Turned a fixed LLM pipeline into a ReAct agent (smolagents) that picks its own tools (Yahoo Finance, SMA50/SMA200 trend, news, weather) with a swappable LLM (Groq, Gemini).
- CI/CD in GitHub Actions: ruff and pytest on Python 3.10–3.12, deploying the Gradio demo to Hugging Face Spaces after CI passes; daily Telegram bulletin via cron.

**Intelligent Game Validation** | *TypeScript, Node.js, Playwright, OCR, GitHub Actions* | GitHub 2026

- 4th place in the Betsson challenge, one of several at Hackathon UMA 2026 (300+ students); wrote all the code of the team's TypeScript/Playwright tool that validates casino games in cross-origin canvas iframes as the player sees them, with no DOM access.
- Built a computer-vision pipeline (pixelmatch, template matching, tesseract.js OCR) cross-checked with provider traffic; CI in GitHub Actions with 42 unit tests and a fault-injection negative control.

**Bancosol Campaigns** | *Java 17, Spring Boot, JPA, Spring Security, JWT, MariaDB* | GitHub 2026

- Implemented the Captain role module of a team Spring Boot web app: dashboard for assigned stores, shifts and volunteers, incident registration, and access limited to each user's own data with Spring Security and JWT.

## EDUCATION

**University of Málaga** 2023 – 2027
*BSc in Software Engineering* Málaga, Spain

**University Extension in Blockchain Technologies (490h)** Feb. 2026 – Jun. 2026
*University of Málaga (UMA)* Málaga, Spain

- Distributed systems, smart contracts (Solidity, Daml), decentralized infrastructure (Canton, gRPC) and blockchain audit fundamentals.

## LANGUAGES

Spanish (native), English (C1, Cambridge).